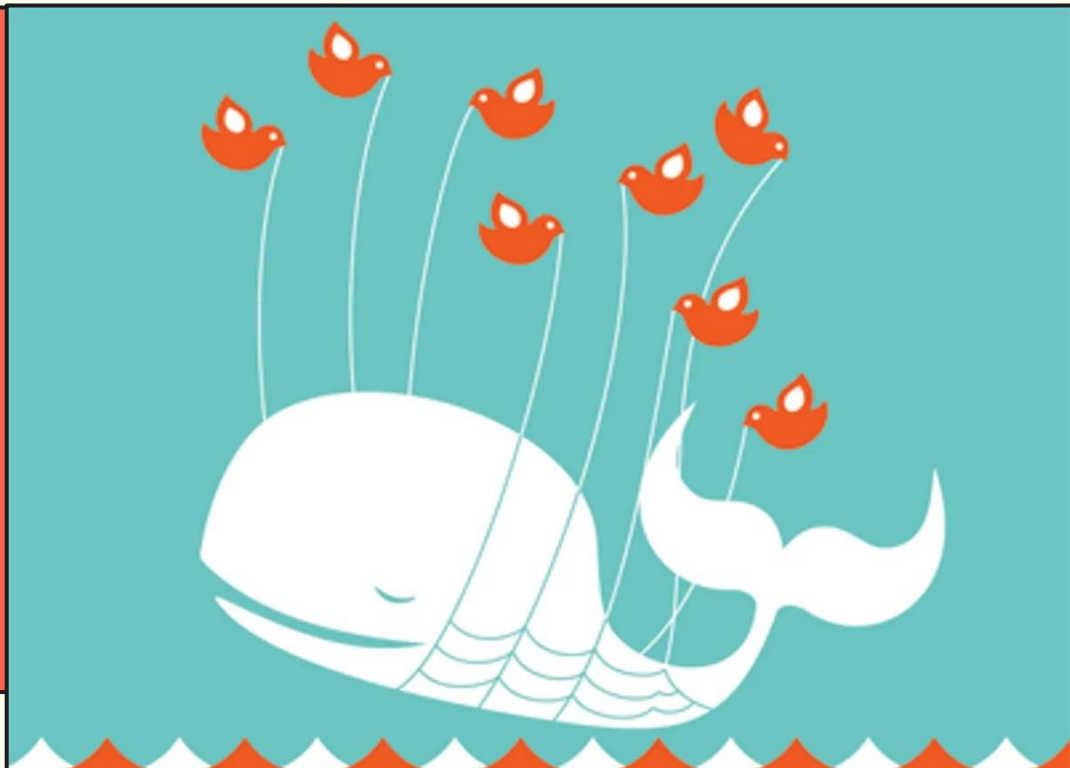


Richard Rose
HUFS / Yonsei

Validating Azure Pronunciation Assessment

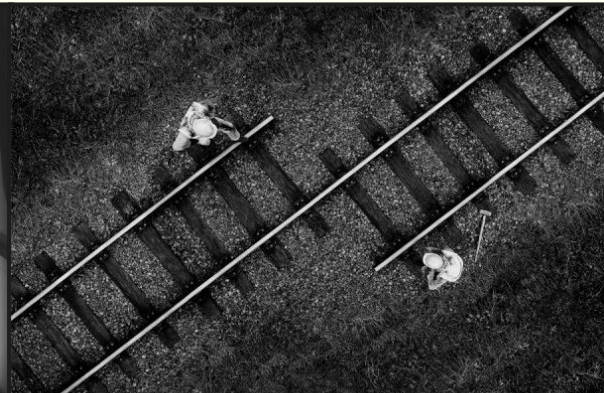
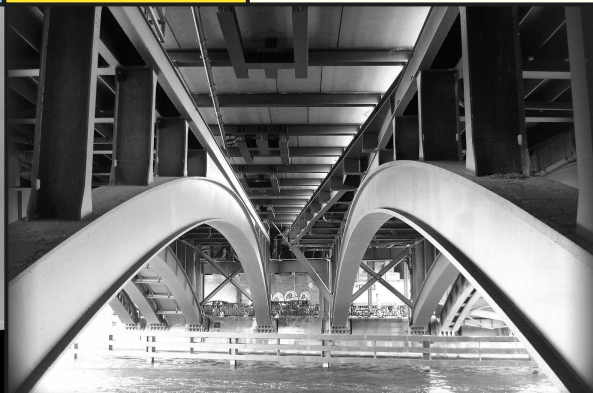
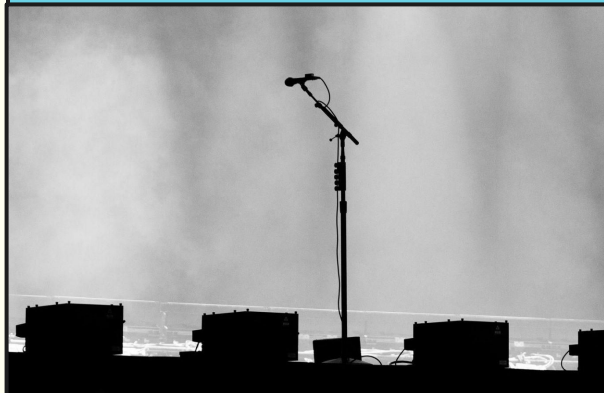
Live Speech Test Demo

- Choose any text
- I'll read aloud and test my speech
- [Demo Link Here!](#)
- Later we can check the analysis PDF
- You can also test your own speech



Presentation Overview

3



The Problem & Potential Solutions

Neural network speech assessment is becoming more prevalent, but validation of these tools remains incomplete.

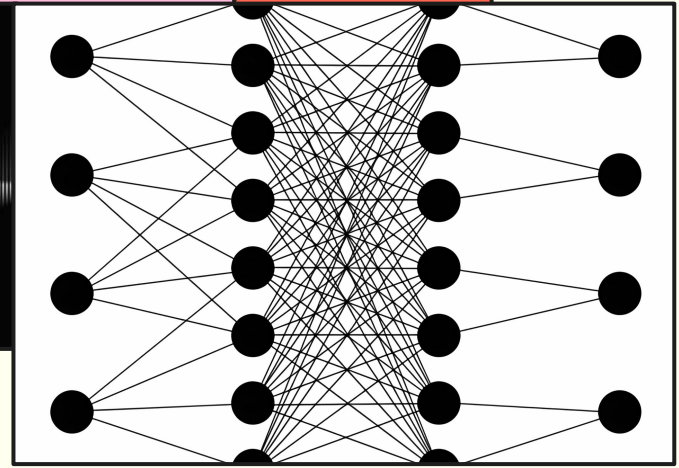
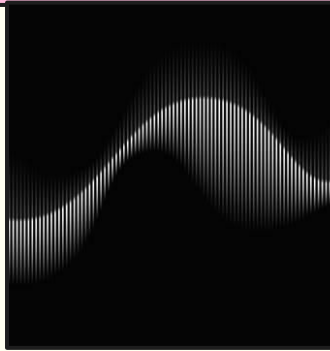
Methods & Measurement

ICNALE corpus ($n = 3,510$)
& classroom data ($n = 66$)
Comparisons of Azure scores with linguistic indices and human ratings

Results & Future Directions

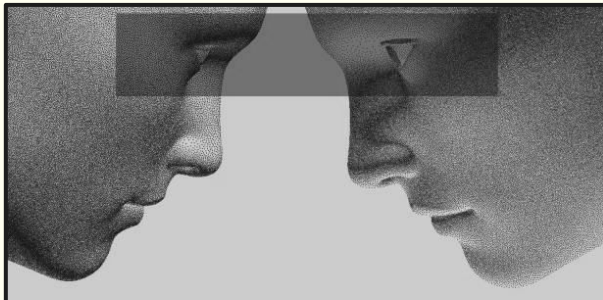
Strong rank-order alignment
but large calibration differences

The Problem & Potential Solutions



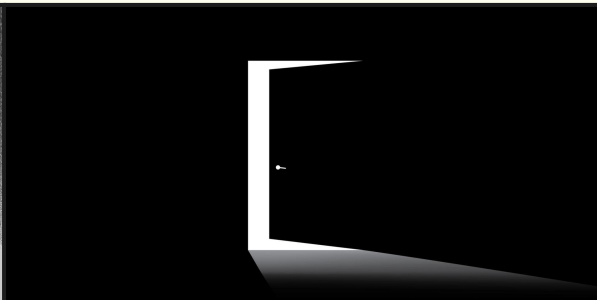
Speech Assessment Challenges

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Interrater variability

- Speech evaluation is inherently subjective
- Trained raters differ in applying scoring criteria
- Performance standards are complex and difficult to operationalize consistently



Limited independent validation

- Largely based on human-machine correlations
- Evidence often produced by test developers
- Rater characteristics and procedures are underreported

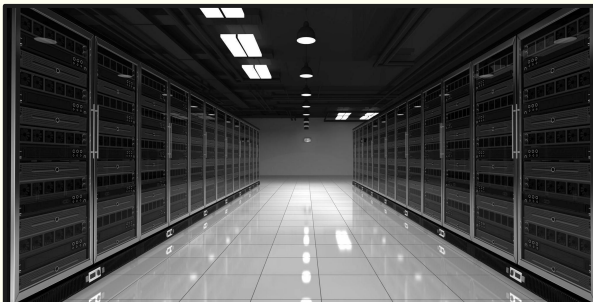


Accessibility constraints

- High cost and dependence on trained raters
- Limited interpretability of scores for teachers and learners

Speech Assessment Solutions

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Commercial automated systems

- Widely deployed (e.g., SpeechRater, Versant, DET)
- Evolved from feature-based models to neural systems
- Scores derived from statistical mappings to human ratings

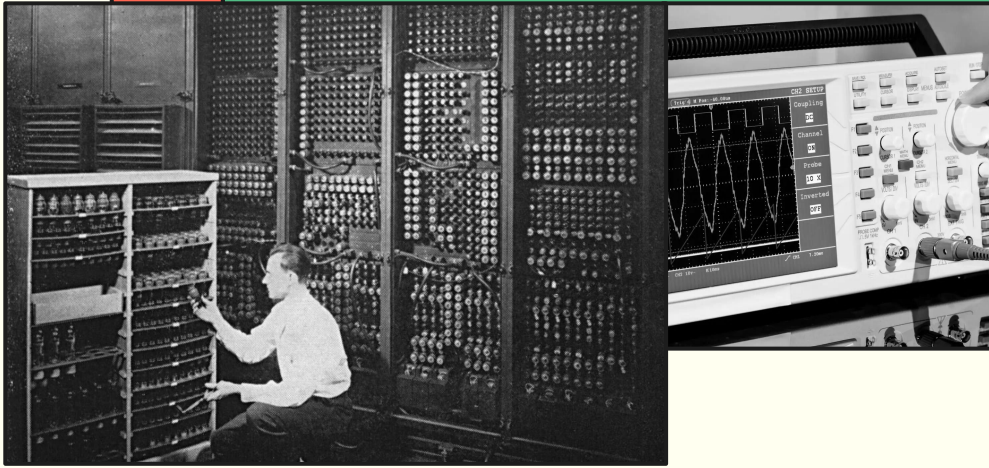
Experimental automated systems

- Machine learning models approximate human speaking judgments
- Use of hierarchical and transformer-based architectures
- Strong predictive performance within datasets

Azure pronunciation assessment

- Production-level system for Accuracy, Fluency, Prosody
- Combines speech recognition features with multi-level modeling of speech patterns
- Evaluated primarily through correlations with expert ratings

Methods & Measurement



The Datasets

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International Corpus Network of Asian Learners of English (ICNALE)

3,510 spoken monologues from learners across 10 countries, with CEFR levels (A2-B2+) assigned using external standardized benchmarks (e.g., TOEFL, TOEIC, IELTS).



Classroom contexts:

University EFL settings in China, Korea, and Japan using a controlled read-aloud task (Please Call Stella) under typical instructional conditions.

Classroom participants:

66 university-level EFL learners from China (28), Korea (20), and Japan (18).

Classroom raters:

Three experienced EFL professionals assigned scores on a calibrated 0-100 scale following a structured calibration procedure.

Linguistic Indices

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Ntypes

Total number of words produced; a baseline measure of speech productivity and output quantity

Ntokens

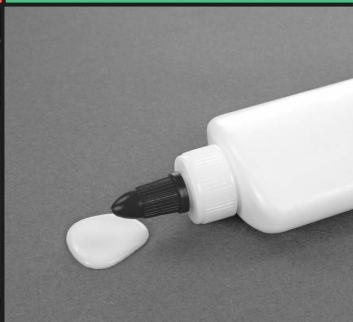
Total number of unique words produced; reflects lexical breadth and vocabulary range within a response.

MATTR

Moving-Average Type-Token Ratio: A text-length-stable measure of lexical diversity calculated across moving windows of speech, reducing inflation from longer responses

MTLD

Measure of Textual Lexical Diversity: A sequential lexical diversity index designed to minimize text-length effects while capturing sustained vocabulary variation across discourse.



Prosody, Accuracy & Fluency

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Prosody

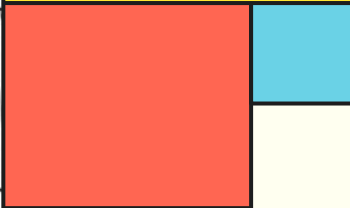
- Stress, rhythm, and intonation
- Helps listeners process meaning

Accuracy

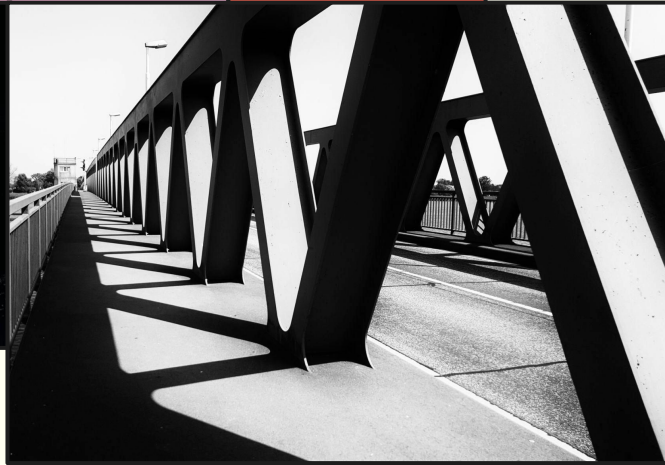
- Segmental control
(vowels, consonants, clusters)
- Contrasts necessary for intelligibility

Fluency

- Temporal control (rate, pauses, repairs)
- Reflects processing efficiency



Results & Future Directions



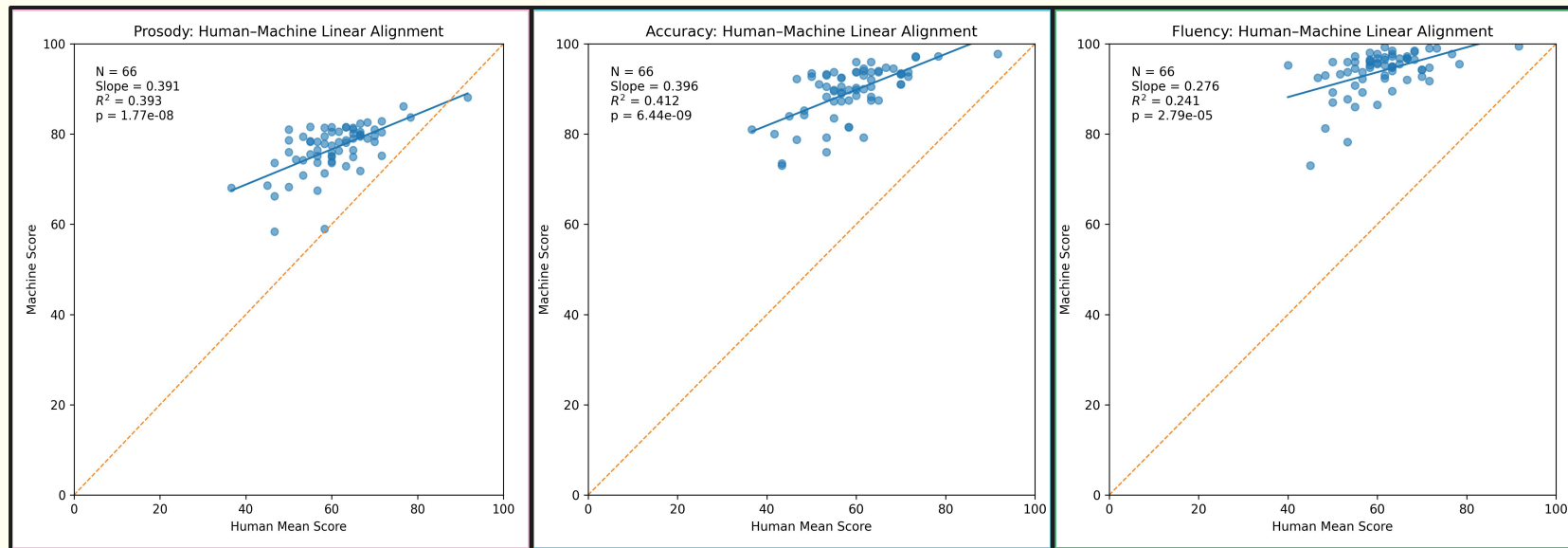
Correlates in Context

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CEFR Correlates	Spearman's ρ	Note:
1. TAALED basic ntypes	0.265	These are Spearman's ρ correlations (all $p < .001$) between CEFR proficiency and selected spoken-task indicators in the ICNALE corpus. Token and type counts are included as reference benchmarks, consistent with prior spoken lexical diversity research (e.g., Kyle et al.), in which MATTR and MTLD are evaluated relative to these foundational measures.
2. AZURE prosody	0.261	
3. AZURE overall pronunciation	0.256	
4. AZURE accuracy	0.242	
5. TAALED basic ntokens	0.241	
6. TAALED MTLD	0.213	
7. TAALED MATTR 50	0.212	
8. AZURE fluency	0.186	

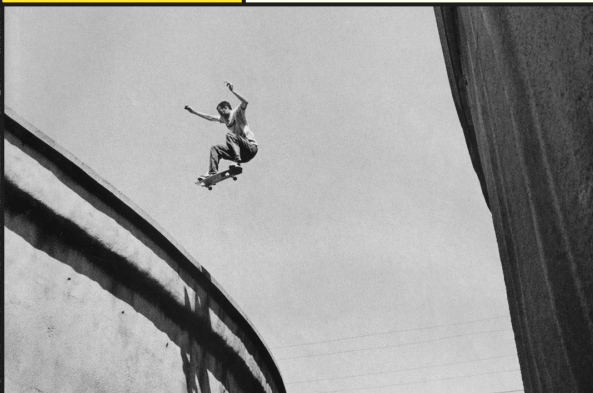
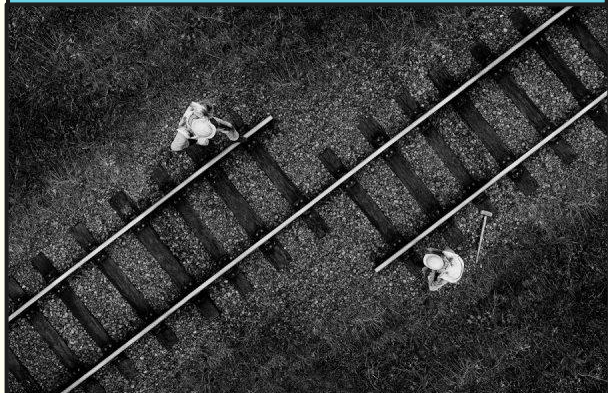
Construct Validation

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Discussion

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Signal, Scale, and Fluency

- **Strong rank-order agreement** (robust re: accuracy & prosody)
- **Systematic bias** (misalignment between human & machine scores)
- **Fluency measurement issues** (rater variability & misalignment)

Limitations & Future Research

- **Monologue task only** (extend to diverse/interactive tasks)
- **Human rating variability** (refine constructs & link scores)

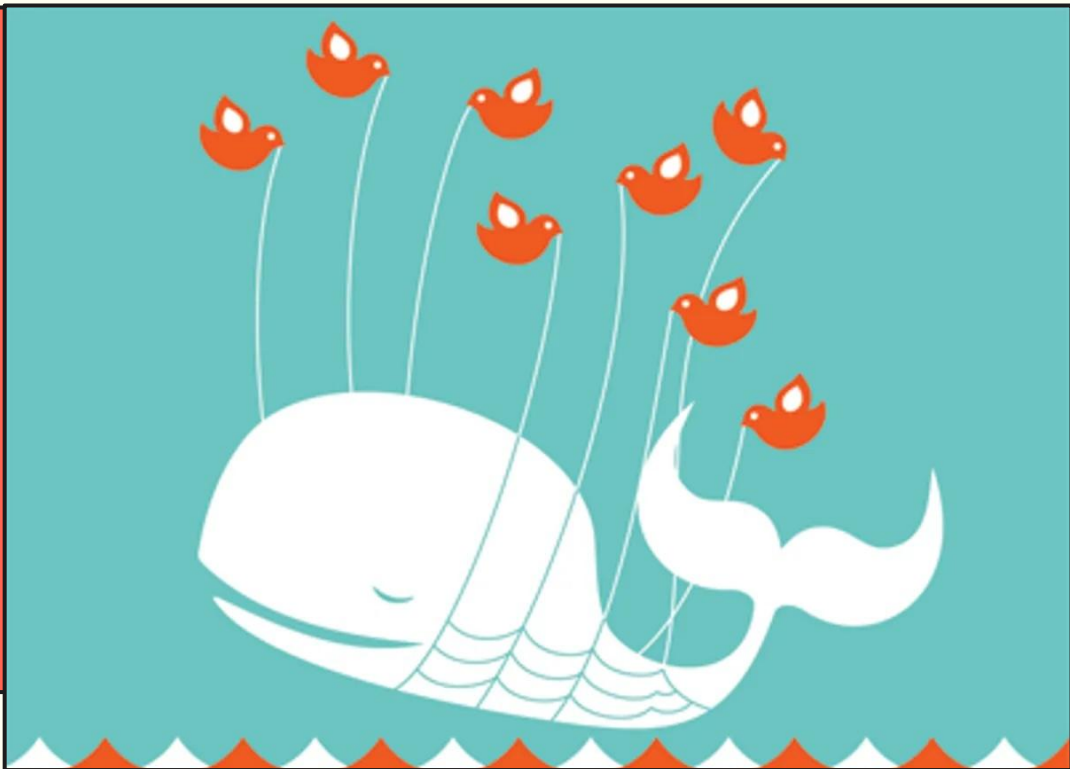
Implications for Practice

- **Useful feedback for formative use**
- **Supports ipsative tracking** (change over time)
- **\$1/hr, 33 languages**
- **Requires calibration or human oversight for high-stakes use**

Live Speech Test Demo

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- Let's check the results of the demo!
- [Demo Link Here!](#)
- [Demo Results Here!](#)
- [Backup Results](#)
(π π)



Q&A

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Citation:

Rose, R., Gaynor, B., & Fowler, J. (2026).
An Argument-Based Validation of
Azure Pronunciation Assessment
Across Corpus and Classroom Data.
Manuscript in press.

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- Zelinski, R. W. (2008). The listener: No longer the silent partner in reduced intelligibility in speech communication. In L. W. Gregg & E. Steinberg (Eds.), *Handbook of speech communication*. San Diego, CA: Sage. <https://doi.org/10.1016/j.specom.2007.11.004>

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